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Detection of on-manifold adversarial attacks via latent space transformation

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ABSTRACT

Out-of-distribution (OOD) generalization is critical for reliable intrusion detection systems (IDS), yet current methods often falter against stealthy, on-manifold adversarial attacks that mimic ID data. To solve this challenge, we propose a semi-supervised approach that applies an invertible transformation to the latent space and leverages changes in differential entropy to detect OOD samples. Experiments on the KDD99 and X-IIoTID datasets demonstrate that our approach outperforms state-of-the-art defenses, providing enhanced robustness and generalizability for IDS.

1. Introduction

Detecting anomalies in network traffic is crucial for maintaining security and ensuring operational integrity in critical infrastructures. These anomalies, when analyzed closely, often align with the characteristics of Out-Of-Distribution (OOD) samples (Corsini and Yang, 2023). Although these anomalies do not necessarily affect the overall data distribution, as evident in Corsini and Yang (2023), they can result in deviations within localized segments of the distribution. This nuanced effect challenges the conventional approaches to anomaly detection, complicating the processes of detection of OOD samples. In the cybersecurity domain, OOD samples are commonly detected using Machine Learning (ML)-based Network Intrusion Detection Systems (NIDS). ML models play a critical role in distinguishing between normal In-Distribution (ID) samples and anomalous OOD samples, enhancing system security and preventing breaches.

Despite achieving State-Of-The-Art (SOTA) performance in many tasks, ML models possess inherent vulnerabilities that can affect their reliability and security (Goodfellow et al., 2014). For example, carefully crafted perturbations — such as bounded noise added to OOD samples — can cause these models to misclassify such inputs as ID (Stutz et al., 2018). Adversarial attackers exploit these vulnerabilities by subtly modifying network traffic patterns to mimic legitimate behavior. For instance, attackers may employ feature-level attack methods (Sarıkaya et al., 2023; Adeke et al., 2025) to alter packet sizes (e.g., slightly reducing large UDP packet sizes in a DDoS attack), tweak the time intervals between packets (thereby avoiding detection based on abnormal burst patterns), or randomize flow durations to deviate from known signatures. In Denial-of-Service (DDoS) scenarios, recent studies have shown that such alterations can reduce the performance of ML-based IDS to 50% (Aiken and Scott-Hayward, 2019). These subtle perturbations complicate detection: they increase False Negatives (FN) by allowing ODD activities to go unnoticed, while overly sensitive models may generate False Positives (FP) by flagging ID yet slightly OOD traffic. Moreover, adversarial attack techniques are highly significant when defenders lack access to source code or need to predict how evolving attack strategies might alter traffic signatures to deceive ML models. In such cases, simulating adversarial scenarios enables researchers and practitioners to anticipate how subtle perturbations can bypass current IDS, which in turn informs the redesign or enhancement of existing ML-based defenses (Al-Fawa'reh et al., 2025). Additionally, employing adversarial attacks as evaluation tools helps uncover critical issues — such as model drift and data snooping — in the IDS domain, ultimately guiding the development of more robust cybersecurity solutions.

Although various defense solutions against such threats do exist (detailed further in Section 2), adversaries continue to develop novel strategies to deceive them. This dynamic often resembles a never-ending cat-and-mouse game. Among these emerging threats are on-manifold attacks, wherein adversarial attacks remain within the normal data manifold, i.e. within the underlying structure or distribution of legitimate data (Xiao et al., 2025). These perturbations, although subtle and blending with expected data distributions, pose significant detection challenges (Zhang et al., 2022; Al-Fawa'reh et al., 2025). Specifically, they do not introduce easily identifiable anomalies or OOD features. On-manifold attacks can be found near the classification boundaries between ID and OOD classes, specifically closer to the ID boundary (Al-Fawa'reh et al., 2025).

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Several approaches have been developed in the literature to detect OOD samples (covered in Section 2). This body of research often concentrates on achieving high-performance metrics using a single test dataset, where the training and testing data share the same underlying distribution (Zou and Liu, 2023). However, very often this assumption does not hold true in real-world applications, highlighting a critical gap. Additionally, the robustness of models against OOD adversarial attacks remains underexplored. Most studies focus on traditional adversarial attacks (such as off-manifold attacks), primarily within computer vision tasks. Recent attacks, such as on-manifold adversarial attacks, have received less attention, particularly in other domains and data types, such as cybersecurity and tabular data. In this paper, we begin by evaluating existing SOTA models in the context of OOD detection, anomaly detection, and adversarial robustness. Our evaluation focuses on model generalization and their performance under on-manifold adversarial scenarios. The results presented in Section 6 (Tables 2, 4 & 5) show that the performance of these models drops significantly. On average, the F1 scores for supervised defenses decline by approximately 21% under adversarial conditions when the attacker has black-box knowledge about the model and defenses compared to their performance on clean data, as demonstrated in our experiments (Section 6.1) using data collected from Industrial IoT networks. These findings indicate that even when OOD generalization algorithms perform well in various scenarios, they remain highly vulnerable to on-manifold adversarial attacks.

Motivated by these limitations, we introduce **E**ntropy-**D**riven **L**atent **T**ransformation (EDLT), a new semi-supervised approach designed to provide a robust and generalizable performance without requiring prior knowledge of OOD and on-manifold adversarial attacks. EDLT focuses on analyzing changes in Differential Entropy(DE) during transformations in the latent space, drawing inspiration from principles in physics.

Physics, with its universal and reliable principles, has always provided a stable foundation for understanding the world around us. By applying these time-tested concepts to our models, we hypothesize that our created solution will be not only more robust to adversarial attacks but also adaptable to a wide range of real-world scenarios. Our goal is to address the gaps in current research with methods that are both grounded in theory and practical in application. We found that energy-based entropy is a promising approach for detecting changes in distribution and identifying outlier samples. Although DE has not been extensively studied for this task, it demonstrated significant potential (as shown in Section 6. However, DE is computationally expensive to utilize with high-dimensional data. Therefore, we first studied it using low-dimensional data representations and utilized density estimation as a proxy to compute DE efficiently using Normalizing Flow (NF). Our approach focuses on studying the change in DE distribution during a series of transformations in the latent space, then establishing a baseline to identify anomalous samples. If the DE value is high, the input samples are likely OOD; otherwise if low, the samples are considered ID. To the best of our knowledge this specific approach has not been investigated in relevant literature.

Our main contributions are summarized as follows:

- We propose a robust and generalizable OOD detection approach, EDLT, which leverages changes in DE within the transformed latent space. EDLT demonstrates superior performance compared to SOTA methods in both robustness and generalizability.[1]
- We extensively evaluate the robustness of EDLT against on-manifold adversarial attacks, comparing it with other SOTA models.
- We assess the generalizability of EDLT by testing it alongside SOTA models on cybersecurity datasets, including the KDD99 dataset (Tavallaee et al., 2009), which targets computer networks, and the X-IIoTID dataset (Al-Hawawreh et al., 2022), which focuses on X-IIoTID networks.

The paper is structured as follows: Section 2 presents an overview of prior work on OOD detection. Section 3 provides foundational background on the core components underlying this research. Section 4 introduces the proposed method, and Section 5 outlines the experimental settings. The results and analysis are presented in Section 6. Section 7 covers computational efforts and comparisons. Section 8 discusses the limitations and Section future directions 9. Followed by an ablation study in Section 10. Finally, the conclusion is presented in Section 11.

2. Related work

Adversarial attacks are increasing rapidly in both frequency and sophistication, making them more realistic and harder to detect. This escalation poses significant risks to safety across critical domains such as autonomous driving, healthcare, and financial services, where robust security measures are essential. Although researchers have proposed various defense approaches, these methods often reduce the accuracy of models when tested on clean data or prove to be completely ineffective, as evidenced in AprilPyone and Kiya (2020) . This reflects a relentless arms race between defenders and attackers, where each advance in attack strategies prompts an urgent need for enhanced defensive techniques. Table 1 provides a summary of existing studies and their limitations, categorized based on key aspects such as the study's approach, advantages, limitations when bypassed by a model, focus on on-manifold adversarial attacks, and whether the proposed defense achieves a satisfactory level of robustness and generalizability. Key studies and their descriptions are provided in the following subsections.

2.1. Adversarial training

This approach relies on generating attack samples and using them to train the model and is considered a SOTA approach (Madry et al., 2017; Wang et al., 2020; Zhang et al., 2019). However, this approach can significantly increase the time required for the training process and may negatively affect the model's accuracy on clean data. Models trained using this approach learn to generalize across a broader range of inputs, including both normal and adversarial, which can dilute their effectiveness on the normal samples they were originally designed to handle best (Bogun et al., 2021; Goodfellow et al., 2014). It also requires access to labeled adversarial samples. An example of this defense is the work presented by Xiao et al. (2025), where the authors proposed an on-manifold adversarial training strategy based on conditional generative models and an eigen space approach. Experiments showed that their approach achieved better performance compared to regular training methods such as (Madry et al., 2017).

2.2. Input prepossessing

This approach focuses on detecting and restoring the malicious parts of an input to produce a normal input or reducing the attack surface . Several notable methods in the literature, such as MagNet (Meng and Chen, 2017), Feature Squeezing (FS) (Xu et al., 2018), and Denoising Autoencoders (DAE) (Sethi et al., 2019; Al-Hawawreh et al., 2021), fall under this category. Additionally, some methods involve using generative models for input transformation (Samangouei et al., 2018), where a GAN learns the distribution of clean images and removes potential adversarial perturbations. While this group of defenses can perform well for specific tasks such as OOD detection in computer vision and NIDS domains, they cannot handle on-manifold attacks, as will be demonstrated in this paper in Section 6.

[1] The code and implementation details can be found at https://github.com/mohdah200/EDLT.

2.3. Post-processing

Post-processing defenses modify a model's decision process *at testing time* without changing training data or model parameters. For example, Cohen et al. (2024) introduced *Tree Test Time Simulation (TTTS)*, a post-processing method designed specifically for decision trees. TTTS employs *Monte Carlo-based decision perturbations* during testing to enhance robustness against adversarial attacks targeting decision paths. Unlike pre-processing or training-time defenses, TTTS operates on a fully trained decision tree without altering its learned parameters or retraining the model. However, while TTTS improves resilience, it *increases testing time* and remains *vulnerable to on-manifold transferability attacks*, as demonstrated in Section 6.

2.4. One class-based OOD detection

This approach focuses on semi-supervised training of models using ID samples and employs a deviation score based on Reconstruction Error (RE) values derived from statistical data. The authors in Al-Fawa'reh et al. (2025) proposed a Data Refinement Module (DRM) based on disentangled and denoising AEs to detect adversarial samples. Although this approach successfully detects the majority of attacks, it lacks generalizability, as demonstrated in the experiments conducted in Section 6.

2.5. Detect and reject

This approach focuses on detecting anomaly samples and rejecting them. Originally created to detect OOD samples, it can also be used to detect adversarial samples. Well-known methods for OOD detection include:

- **Confidence-based OOD Detection :** This method operates by utilizing a model's prediction certainty as an indicator of whether data is ID, assigning lower confidence scores to OOD samples (Hendrycks and Gimpel, 2016). However, confidence levels alone may not always be reliable indicators of OOD data, as some studies have indicated (Corsini and Yang, 2023). Furthermore, in the case of unsupervised OOD detection, no labels are provided to the model, and OOD samples are encountered only during the testing time. This suggests that this defense approach is impractical.
- **Reconstruction-based OOD Detection :** This method operates on the assumption that consistently high reconstruction error (RE) values between input and output samples indicate OOD data. A recent study by Sarıkaya et al. (2023) introduced the RAIDS model to enhance the robustness of IDS against adversarial attacks, employing RE in the detection process. Although the authors demonstrated its resilience against generative adversarial attacks, we successfully bypassed it using on-manifold attacks. The reason behind this is that on-manifold adversarial examples remain within the learned data distribution, leading to low RE values, making them harder to detect by reconstruction-based methods.
- **OOD-Model Selection (OOD-MS) :** OOD-MS dynamically selects the most suitable classifier for each input, improving robustness against adversarial and OOD samples. Instead of relying on a fixed model or confidence scores, it leverages adaptive selection techniques (e.g., Multi-Armed Bandits) to optimize detection and prevent attackers from exploiting static decision boundaries. For example, the author in Paya et al. (2024) introduces Apollon, which enhances OOD detection and adversarial robustness by dynamically selecting the most effective classifier for each input using Multi-Armed Bandits (MAB) with Thompson Sampling, making it adaptive against evolving attacks. Similarly, the study in Barik and Misra (2024) presents a preprocessing approach that reduces the attack surface using z-score normalization and Singular Value Decomposition (SVD), followed by an optimization algorithm for classifier selection. However, both models lack robustness against on-manifold attacks, as demonstrated in Section 6, making them more vulnerable to adversarial manipulations within the learned data distribution.
- **Distance-based OOD Detection :** This method relies on the spatial relationship between data points (Lee et al., 2018). In this method, prototypes or centroids of ID samples are established, and distances from these points are calculated using metrics such as Mahalanobis (Sun et al., 2022), Euclidean (Sun et al., 2022), or Cosine distance (Amirullah et al., 2020). Samples that lie significantly far from any class prototype are flagged as OOD. This method is not examined in this paper as it can easily be fooled by some attacks, such as Deepfool (Moosavi-Dezfooli et al., 2015).
- **Density-based OOD Detection :** This method focuses on modeling the distribution of ID samples, using either raw or latent representation (Yang et al., 2021). This method identifies OOD samples by detecting those located in low-density regions of the data distribution. Techniques include parametric models, such as Mixture of Gaussians, and non-parametric approaches, such as Kernel Density Estimation, which offer robust methods to understand and characterize the data landscape (Shah et al., 2023). Despite the challenges highlighted in Corsini and Yang (2023) such as the need to validate a prior assumption and scalability issues, we adopt a density-based approach in this paper. We leverage recent advancements in generative AI models, namely NF which are used as precise density estimators. Although some earlier and recent studies, such as Zhang et al. (2021), Kirichenko et al. (2020), Gudovskiy et al. (2022), Rudolph et al. (2021), Cho et al. (2020) and Cook et al. (2024), have explored the density estimation approach, these studies still exhibit limitations in investigating the robustness and generalizability of the proposed models. Notably, authors in Zhang et al. (2021) and Kirichenko et al. (2020) have advised against using density estimation for OOD detection, arguing that it captures only the low-level statistics of the distribution. Conversely, authors in Gudovskiy et al. (2022) and Rudolph et al. (2021) applied density estimation to image segmentation tasks and observed promising results, however they do acknowledge its limited generalizability to other domains, as evident in Cho et al. (2020). Similarly, Cook et al. (2024) explored the application of the Glow NF model but found it did not generalize well, as evidenced in Cho et al. (2020). Moreover, Cho et al. (2020) and Rudolph et al. (2020) investigated its use in image classification tasks but did not assess the robustness against adversarial attacks. They also utilized this approach as an auxiliary model for classification tasks, potentially impacting overall performance.

A major gap identified in the literature, and reinforced by our evaluation in this paper, is the absence of a model that is both generalizable and sufficiently robust to effectively detect OOD samples and on-manifold adversarial attacks (stealth OOD samples) simultaneously. Existing models often over-rely on prior knowledge about the attacks, such as labeled attack samples, which limits their effectiveness in real-world scenarios. Models that rely on RE, such as VAEs, are particularly prone to this limitation, as demonstrated in Section 6. Similarly, models that depend on density estimation, such as NF, struggle to simultaneously guarantee generalizability and robustness, as shown in the experimental Section 6. Motivated by prior works such as Al-Fawa'reh et al. (2025), Tarchoun et al. (2023) and Rudolph et al. (2020), we adopt a semi-supervised learning approach to address these challenges. Specifically, we focus on employing density estimation as a proxy to derive anomaly scores, with the density estimation relying on the log-likelihood value and investigating characteristics such as density and changes in DE within the latent space of ID samples during a series of transformations. To the best of our knowledge, this approach has never been investigated before, at least in the context of cybersecurity.

Table 1
Summary of adversarial attack defense methods in literature.

Study	Approach	Advantages	Limitations	Bypassed by	On-Manifold?	R & G?
Xiao et al. (2025)	On-Manifold Adv. Train	Considers manifold; better vs. standard training	High computation; needs generated adversarial samples	Attacks outside manifold assumptions	✓	✓
Meng and Chen (2017)	Input preprocessing	Plug-in; filters inputs	Reduction in clean accuracy	On-manifold attack	✗	✗
Barik and Misra (2024) and Xu et al. (2018)	Input preprocessing	Reduce the attack surface	Reduction in clean accuracy	On-manifold attack	✗	✗
Sethi et al. (2019) and Al-Hawawreh et al. (2021)	Input preprocessing	Removes minor noise	Reduction in clean accuracy	AE-targeted adversaries	✗	✗
Cohen et al. (2024)	Post-Preprocessing	Simple; Fast	Increases the testing time	On-manifold attack	✗	✗
Al-Fawa'reh et al. (2025)	One-Class OOD	Relies solely on ID samples; Good for off-manifold adv attacks	RE unreliable	AE-targeted adversaries	✗	✗
Hendrycks and Gimpel (2016)	Detect & Reject	Simple; Baseline for OOD	Not reliable	Logit manipulation	✗	✗
Sarıkaya et al. (2023)	Detect & Reject (RE)	Simple	RE unreliable	On-manifold attack	✗	✗
Paya et al. (2024) and Barik and Misra (2024)	Detect & Reject (OOD-MS)	Dynamically select the best model for OOD	Prone to attack transferability	On-manifold attack	✓	✗
Lee et al. (2018)	Detect & Reject (Distance)	Simple & common for OOD	Relies only on n-norm	Attacks outside distance assumptions	✗	✗
Zhang et al. (2021)	Detect & Reject (Density)	Good for OOD detection	Needs to validate a prior assumption	On-manifold stealth attacks	✗	✗

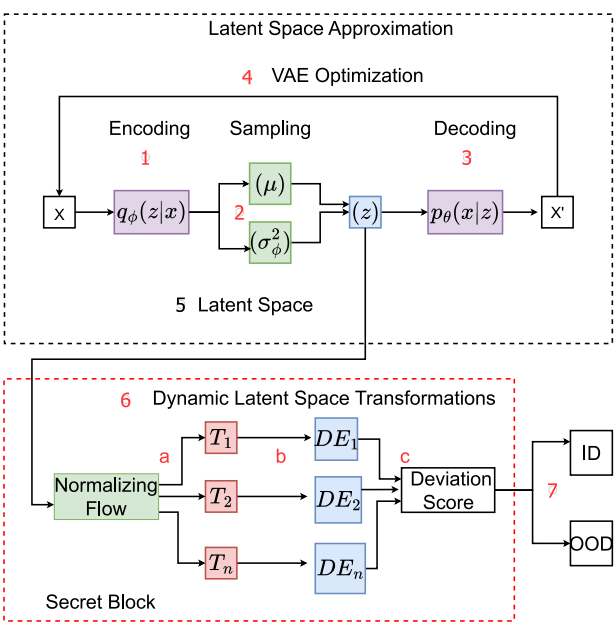

Fig. 1. The proposed approach.

3. Background

3.1. Information theory

Information theory provides a foundational framework for analyzing data through entropy, a concept that quantifies uncertainty or randomness within a system. By leveraging entropy and DE, we can better understand data density and variability, which are valuable for identifying anomalies. By analyzing the randomness based on the probability density functions of data distributions, we can systematically identify deviations that may indicate OOD. Entropy was initially formulated by Shannon (1948). For discrete variables, it measures the expected amount of information needed to identify a random variable's state. When dealing with continuous data, entropy evolves into DE, which assesses the entropy within a continuous probability density function $f(x)$ as given in Eq. (1) (Shannon, 1948).

$$H(X) = -\int f(x)\log f(x)\,dx \tag{1}$$

Inspired by recent work presented by Tarchoun et al. (2023), where it was found that some adversarial patches contain a statistically higher amount of information compared to natural data, we employ DE based on density estimation according to Shannon (1948). We hypothesize that this will serve as a crucial indicator of potential adversarial attacks. While Tarchoun et al. (2023) utilized DE to detect adversarial regions in images, it neither addressed on-manifold attacks nor considered transformations of the latent space. Moreover, when we applied their approach to OOD samples, we found that their model performed poorly. T**o the best of our knowledge**, this is the first study to employ DE alongside ID latent space transformations to detect OOD samples.

3.2. Normalizing flow

Normalizing Flow (NF) is a class of probabilistic models that defines a series of invertible transformations to model complex probability distributions (Kobyzev et al., 2019), making them particularly useful for generative modeling tasks. The process starts with a simple, easy-to-sample distribution $q_0(z)$, often a standard multivariate Gaussian $q_0(z) = \mathcal{N}(z; 0, I)$. Then, a NF model applies a series of invertible transformations f_i, each with an associated inverse f_i^{-1} and a tractable Jacobian determinant $\left|\det J_{f_i}(z)\right|$, to the base distribution (Kobyzev et al., 2019). If we have a sequence of transformations $f_1, f_2, \ldots, f_K$, a sample z_0 from the base distribution q_0 is transformed through the flow to get z_K as shown in Eq. (2) (Dinh et al., 2016).

$$z_K = f_K \circ f_{K-1} \circ \ldots \circ f_1(z_0) \tag{2}$$

For each transformation f_i, we can use the change of variables formula to compute the probability density of z_i after applying the transformation as in Eq. (3).

$$q_i(z_i) = q_{i-1}(f_i^{-1}(z_i)) \left|\det J_{f_i^{-1}}(z_i)\right| \tag{3}$$

where $J_{f_i^{-1}}(z_i)$ is the Jacobian of the inverse of f_i at z_i. After applying all K transformations, the density of z_K is given by Eq. (4).

$$q_K(z_K) = q_0(f_1^{-1} \circ \ldots \circ f_K^{-1}(z_K)) \times \prod_{i=1}^{K} \left|\det J_{f_i^{-1}}(f_{i+1}^{-1} \circ \ldots \circ f_K^{-1}(z_K))\right| \tag{4}$$

When selecting an appropriate NF model to study the latent space transformation of ID samples, several notable NF models that perform bijective transformations should be considered to construct complex mappings between probability distributions. These include Planar (Rezende and Mohamed, 2015), Autoregressive Flows (Kingma et al., 2016), Glow (Kingma and Dhariwal, 2018), and RealNVP (Dinh et al., 2016). Key factors such as the complexity of the target distribution, the dimensionality of the data, sampling speed, computational resources, and the proven success of the NF models in various domains are critical in the selection process. In this paper, we adopt a design similar to that of Dinh et al. (2016) and Kingma and Dhariwal (2018), enhanced with a dynamic clamping mechanism.

3.3. Background to adversarial attacks

Adversarial attacks involve introducing carefully crafted perturbations to input data with the goal of causing a ML model to misclassify. Formally, consider a classification function $f : \mathcal{X} \to \{1, 2, \ldots, K\}$, where $\mathcal{X} \subseteq \mathbb{R}^d$ is the input space (e.g., feature vectors derived from network traffic) and K is the number of classes (e.g., benign vs. multiple types of malicious traffic). Given an original sample $\mathbf{x} \in \mathcal{X}$ with ground-truth label y, an adversarial attack seeks a small perturbation $\boldsymbol{\delta}$ such that the modified input $\mathbf{x} + \boldsymbol{\delta}$ is misclassified: $f(\mathbf{x} + \boldsymbol{\delta}) \neq y$, under the constraint that $\|\boldsymbol{\delta}\|_p$ remains small for some norm p (often $p \in \{1, 2, \infty\}$). In *targeted* attacks, the objective is even more specific, forcing $f(\mathbf{x} + \boldsymbol{\delta}) = t$ for a chosen target class $t \neq y$.

In the context of IDS, adversarial perturbations must maintain the "manifold" of legitimate network traffic. That is, despite changes to packet sizes, inter-arrival times, or flow statistics, the resulting adversarial traffic should still appear valid and evade traditional detection rules (e.g., signature-based or anomaly-based detectors). Ensuring this plausibility is essential for real-world attackers who must avoid raising suspicion. Below, we summarize three widely used gradient-based attacks, adapted here to the network intrusion domain, followed by a discussion of generative adversarial attacks that leverage generative models.

- **Basic Iterative Method (BIM) :** BIM is an iterative extension of the Fast Gradient Sign Method (FGSM). At each iteration, a small step is taken along the gradient of the classification loss $\mathcal{L}\big(f(\mathbf{x}), y\big)$, accumulating perturbations (see Eq. (5)) (Kurakin et al., 2017).

$$\boldsymbol{\delta}_{\text{iter+1}} = \text{Clip}_{\epsilon}\left\{\boldsymbol{\delta}_{\text{iter}} + \alpha\, \text{sign}\big(\nabla_{\mathbf{x}} \mathcal{L}\big(f(\mathbf{x}), y\big)\big)\right\} \tag{5}$$

  where α is a small step size, and $\text{Clip}_{\epsilon}\{\cdot\}$ ensures $\|\boldsymbol{\delta}\|_\infty \leq \epsilon$.
- **DeepFool (DF) :** DF aims to find the minimal perturbation needed to cross the nearest decision boundary. It iteratively projects the sample onto the linearized boundaries of the classifier (Moosavi-Dezfooli et al., 2015), refining the perturbation until misclassification is achieved (see Eqs. (6) and (7)).

$$\boldsymbol{\delta} = \arg\min_{\boldsymbol{\delta}} \|\boldsymbol{\delta}\|_2 \tag{6}$$

$$\text{subject to} \quad f(\mathbf{x} + \boldsymbol{\delta}) \neq y \tag{7}$$

- **Carlini & Wagner L_∞ (CW_{L_∞}) :** This attack formulates adversarial example generation as an optimization problem with a custom loss function (see Eq. (8)) (Carlini and Wagner, 2016).

$$\boldsymbol{\delta} = \arg\min_{\boldsymbol{\delta}} \|\boldsymbol{\delta}\|_\infty + c\, \mathcal{L}_{\text{CW}}\big(f(\mathbf{x} + \boldsymbol{\delta}), y\big) \tag{8}$$

  where $c > 0$ balances the perturbation size and the misclassification objective $\mathcal{L}_{\text{CW}}$.
- **Generative Adversarial Attacks using Generative Models :** This approach leverages generative models—such as Generative Adversarial Networks (GANs) or Autoencoders (AEs)—to learn the distribution of legitimate network traffic and produce adversarial samples that lie on (or very close to) this manifold. Conceptually, a generative model $G(\mathbf{z})$ (where $\mathbf{z}$ is a latent vector) is trained to generate "ID-looking" data. To craft adversarial attacks, $\mathbf{x}^{\text{adv}} = G(\mathbf{z})$, the attacker may solve:

$$\mathbf{z}^* = \arg\min_{\mathbf{z}} \Big[\mathcal{L}\big(f(G(\mathbf{z})), y\big) + \lambda\, d\big(G(\mathbf{z}), \mathbf{x}\big)\Big] \tag{9}$$

  where λ is a hyperparameter and $d(\cdot, \cdot)$ is a distance metric measuring how closely the generated sample resembles a real (or specific original) network flow. The adversary thus obtains $\mathbf{x}^{\text{adv}} = G(\mathbf{z}^*)$, which looks highly authentic yet is misclassified by f.

This paper focuses on on-manifold adversarial attacks, a category that has received comparatively less attention in the literature. These attacks are particularly challenging as they ensure that adversarial samples remain within the data manifold, making detection more difficult. A detailed discussion of this attack type, along with methods for generating adversarial samples that preserve realism, is provided in Section 5.4.

4. Method

The overall pipeline for detecting OOD samples and on-manifold attacks is illustrated in Fig. 1. It begins with extracting low-dimensional data using a VAE, which is composed of an Encoder (E), a Decoder (D), and a latent space. The encoder maps input data to a latent space by estimating the parameters of a probability distribution (Gaussian), while the decoder reconstructs the input data from samples drawn from this distribution, based on the manifold theory. Next, invertible transformations of the latent space are applied using NF (Step 6 in Fig. 1), ensuring diversity in the representation of ID data. More details about the transformation process are shown in Fig. 2. Finally, changes in DE are studied to determine the deviation score for identifying ID samples (Steps 6a, 6b & 7 in Fig. 1).

4.1. Latent space approximation

In this phase, we leverage the principles of manifold learning to move beyond raw high-dimensional inputs and instead operate within a latent space where data resides on a smoother, lower-dimensional manifold (Friedl et al., 2019). Learning from such manifold-structured representations enhances both learning efficiency and model interpretability, compared to attempting to capture patterns directly in high-dimensional input space (Xiao et al., 2025). To this end, we employ a VAE, a generative model well-suited for manifold learning due to its ability to infer complex underlying distributions and disentangle essential data features (Doersch, 2016). The VAE's encoder compresses input data into a latent space governed by a Gaussian distribution defined by μ and σ_ϕ^2 (Steps 1 & 2 in Fig. 1), while its decoder reconstructs the original samples from these latent representations. Critically, the training objective balances reconstruction quality with a regularization

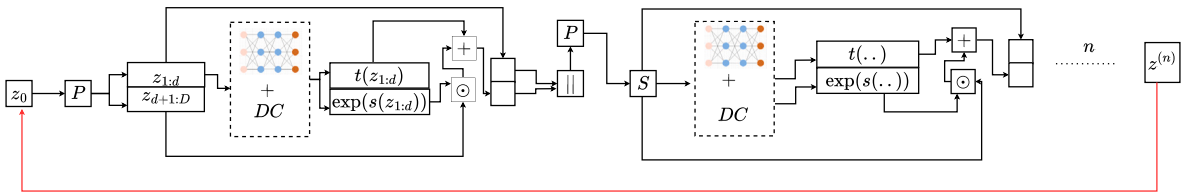

Fig. 2. A detailed view of the latent space transformations illustrating the sequence of operations including Permutations (P), affine coupling layers, flow Concatenation (||), Splitting (S) and Dynamic Clamping (DC).

term that nudges the latent space toward a well-behaved, standardized prior distribution (Step 4 in Fig. 1). By applying the Reparameterization Trick ($z = \mu + \sigma_\phi^2 \cdot \epsilon$ with $\epsilon \sim \mathcal{N}(0,1)$; Step 2 in Fig. 1), we enable efficient gradient-based optimization through stochastic sampling (Step 4 in Fig. 1). As a result, the manifold-based latent representation obtained in Step 3 in Fig. 1 not only streamlines downstream tasks, such as OOD detection, but also provides a more robust foundation for recognizing subtle patterns inherent to ID data, thereby mitigating the pitfalls of purely high-dimensional approaches.

4.2. Dynamic transformations and clamping mechanism

In this phase, we center our investigation on the latent space transformations and their relationship to the DE of the resulting distributions. Fig. 2 illustrates the on-manifold latent space transformations, composed of permutations, affine coupling layers, and dynamic clamping mechanisms. In the forward direction (black arrows in Fig. 2), input data is processed through these stages: permutations rearrange features, affine coupling layers apply scale and shift transformations conditioned on subsets of the data, and dynamic clamping regulates values to ensure stability and robustness. In the backward direction (red arrow in Fig. 2), the transformations are bijective, allowing us to estimate the Probability Density of the input data by enabling the exact recovery of its original distribution.

By examining how key-driven NF transformations alter the latent space, we gain insights into how the complexity and uncertainty (as measured by DE) evolve in response to adaptive defense strategies. This approach not only refines our understanding of the underlying geometry and information content of the latent space, but also prepares us for dealing with an elevated threat model, where attackers attempt more sophisticated forms of testing and manipulation. To counter these advanced adversaries, we incorporate the cryptographic principles outlined in Taran et al. (2018). Even though the attacker is aware of the use of permutation-based NF transformations, the exact keys, internal states, and dynamic clamping mechanisms remain hidden. Step 6 in Fig. 1 highlights these mechanisms, which are further detailed in the following subsections. These secret, data-independent transformations continually reconfigure the latent space, ensuring that the distribution is not only intractable from the attacker's perspective, but also maintains high DE that complicates any direct inversion or approximation attempts. The dynamic clamping mechanism further strengthens this defense by continuously reshaping the latent manifold. This adaptive process raises the stakes for the attacker, who faces a non-stationary, high-entropy latent landscape devoid of exploitable patterns. Meanwhile, the defender, with full knowledge of the secret keys and transformations, can reliably reproduce these changes at testing time, maintaining an enduring information advantage. By scrutinizing these transformations in terms of DE and introducing cryptographic resilience measures, we forge a strategic environment in which the latent space becomes an active, robust line of defense. Ultimately, this phase lays the groundwork for a more profound understanding of how intricate, key-secured latent transformations resist even the most determined adversarial threats.

4.2.1. Permutation phase

In this phase, the approximated latent space is transformed through a NF composed of affine coupling layers, as illustrated in Fig. 2. Each layer applies a carefully crafted transformation to enrich the complexity of the latent representation. At the onset of each layer's transformation, a permutation of the latent dimensions is introduced. While this permutation could, in principle, be optimized to preserve semantic relationships between features, we employ a simple random permutation approach here. Notably, this permutation, highlighted by the red box in Fig. 1, can be regarded as a form of a secret key—securely established during training and maintained consistently at testing time—ensuring that only the intended parties can reproduce the exact latent transformations. Our primary focus is not on refining the permutation itself, but rather on showcasing how these latent transformations lead to richer, more expressive distributions that ultimately enhance the model's robustness and adaptability.

4.2.2. Post-permutation phase

In this phase, we divide the permuted ID latent space into two subsets, as shown in Fig. 2, following an approach similar to that described in Rudolph et al. (2020). The first subset, $z_{1:d}$, remains unaltered during this step and is fed into a shallow Neural Network (NN) comprising two layers, chosen for their computational efficiency. This NN outputs parameters $s(z_{1:d})$ and $t(z_{1:d})$ used for scaling and translating the second subset of the latent space, $z_{d+1:D}$, effectively modifying its distribution. The transformation is mathematically expressed as shown in Eq. (10):

$$z_{d+1:D} = z_{d+1:D} \odot \exp(s(z_{1:d})) + t(z_{1:d}), \tag{10}$$

where $\odot$ denotes element-wise multiplication. The functions s and t dynamically adjust the scale and shift of the data features based on the unmodified part of the latent space, thereby morphing the original distribution into a more complex one. To ensure that all dimensions of the latent space are transformed and to increase the complexity of the representation, this process is repeated iteratively, typically three times as determined through empirical tuning. Between each affine coupling transformation f_i, permutation operations σ_i are applied to rearrange the dimensions of the latent space. This sequence ensures that every dimension is affected over the course of the transformations. The iterative process can be represented as:

$$z^{(3)} = f_3 \circ \sigma_2 \circ f_2 \circ \sigma_1 \circ f_1(z^{(0)}), \tag{11}$$

where $z^{(0)}$ is the initial latent representation.

4.3. Dynamic clamping mechanism

To enhance the effectiveness of these transformations and maintain computational stability, we incorporate a dynamic clamping mechanism. This mechanism involves adjusting the NN output parameters $s(z_{1:d})$ and $t(z_{1:d})$ based on real-time feedback from the model's performance. Specifically, we clamp the scaling parameters $s(z_{1:d})$ within learnable bounds $\theta_{\min}$ and $\theta_{\max}$, refining the scaling and translation operations adaptively to new data. This operation is integrated into the network's forward pass, allowing backpropagation to adjust

these clamping parameters based on the loss gradient. The clamped transformation is expressed as shown in Eq. (12).

$$z_{d+1:D} = z_{d+1:D} \odot \exp(\text{clamp}(s(z_{1:d}), \theta_{\min}, \theta_{\max})) + t(z_{1:d}) \tag{12}$$

By dynamically adjusting the scaling factors, we prevent extreme transformations that could lead to numerical instability, ensuring that the model remains robust and efficient. This dynamic clamping mechanism allows the transformations to adapt to new data while maintaining control over the magnitude of the adjustments, ultimately enhancing the model's ability to capture complex data distributions and resist adversarial perturbations.

4.4. Training phase and DE optimization

The proposed model is trained by using gradient descent over the latent representations to optimize the density estimation and the DE $H(z)$. The gradient descent updates are formalized as in Eq. (13).

$$Z_0 = Z,\ Z_{N+1} \quad = Z_N - \alpha \cdot \nabla_Z H(q(Z), Z) \tag{13}$$

where Z is the set of latent representations being transformed by the NF model, $H(q(Z))$ denotes the DE of Z within the latent space, and α is the learning rate. To optimize the DE of the transformed latent space of the ID samples, we need to estimate the density of the latent space. To achieve this, we employ the log-likelihood loss function during the training process of the NF model (Kingma and Dhariwal, 2018). The loss function in our case will be as in Eq. (14).

$$\mathcal{L} = -\frac{1}{N}\sum_{j=1}^{N}\left[-\frac{1}{2}\log(2\pi\sigma^2) - \frac{(z_0^{(j)} - \mu)^2}{2\sigma^2} - \sum_{i=1}^{K}\log\left|\det \mathbf{J}_{f_i}\left(z_{i-1}^{(j)}\right)\right|\right] \tag{14}$$

Here, J_{f_i} represents the Jacobian of the transformation function f_i, which details how the transformation scales the volume elements of the latent space. Maximizing this log-likelihood function effectively trains the NF to fit the data distribution as closely as possible, thus enhancing the fidelity of the generated or transformed samples.

4.5. Differential entropy estimation

To optimize the DE across transformations in the NF model, we analyze how each transformation affects the DE of the latent space. The DE for a continuous random variable z with density $q(z)$ is defined as shown in Eq. (15) (Dinh et al., 2016).

$$H(z) = -\int q(z)\log q(z)\,dz \tag{15}$$

When an invertible transformation f is applied to z, the density $q(z')$ of the transformed variable adjusts according to the change of variables formula (Dinh et al., 2016):

$$q(z) = q(z)\left|\det \mathbf{J}_f(z)\right|^{-1} \tag{16}$$

where $\mathbf{J}_f(z)$ is the Jacobian matrix of the transformation. This Jacobian determinant captures how volume elements in the latent space scale under f, and its logarithm contributes directly to the DE. Thus, the change in DE resulting from f is:

$$\Delta H = \mathbb{E}_z\left[\log\left|\det \mathbf{J}_f(z)\right|\right] \tag{17}$$

Each transformation in the NF model generally affects only part of the vector z, with the other part passing through unchanged. For example, affine coupling layers modify a subset of z based on scaling and translation functions $s(z)$ and $t(z)$, where the Jacobian matrix is structured to allow efficient computation of its determinant. The determinant is simplified as a sum of the scaling function outputs, making it computationally feasible to track DE changes across layers. After applying a series of transformations, the overall DE for the transformed representation $z^{(K)}$ becomes:

$$H(z^{(K)}) = H(z^{(0)}) + \sum_{i=1}^{K}\Delta H_i \tag{18}$$

This cumulative DE calculation allows the model to capture the complex structure of the data distribution effectively. By maximizing the log-likelihood of the transformed samples during training, the NF model indirectly adjusts the DE of the distribution. This approach enables the model to capture both data complexity and unpredictability, making DE a powerful metric for anomaly detection. Significant deviations in DE can highlight OOD or adversarial samples, ensuring that the model remains robust and sensitive to anomalies. The training process optimizes DE to reflect inherent data complexity, facilitating the detection of anomalies and adversarial perturbations within the latent space.

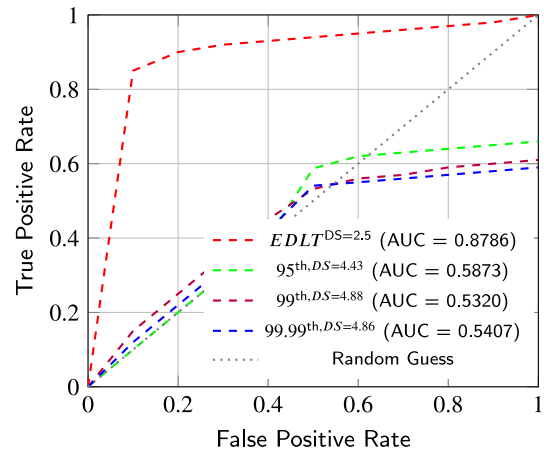

Fig. 3. AUC-ROC scores at different thresholds for the X-IIoTID dataset.

4.6. Deviation score

To identify samples that deviate from their expected distribution, we propose an anomaly Deviation Score (DS) based on the DE of the transformations in the latent space of ID samples. Specifically, we use the DE from the third transformation step, denoted as $H(z^{(3)})$. The DS is calculated using the formula $\text{DS} = \mu_{H(z^{(3)})} + 2\sigma_{H(z^{(3)})} + t$, where $\mu_{H(z^{(3)})}$ is the mean of the DE values of $z^{(3)}$, $\sigma_{H(z^{(3)})}$ is the standard deviation of these values, and t is a tuning parameter, adjustable within ± 5 based on our experimental results and AUC-ROC analysis. Fig. 3 illustrates the AUC-ROC performance across different threshold values for the X-IIoTID using our DS and the well-known percentile threshold from Goodge et al. (2020). EDLT achieves its highest performance when DS is set to 2.5—a setting that also yields the best results for the KDD99 dataset.

5. Experiments

We initially experimented with both joint and disjoint training of the VAE and the NF models, finding that disjoint training yielded more stable and effective results. In the chosen approach, we first train the VAE, which features an encoder with three fully connected layers: input dimension to 128, 128 to 64, and 64 to 12. The VAE uses a 78/121-dimensional latent space parameterized by separate mean and log-variance layers. The decoder mirrors the encoder's structure, using ReLU activations in the hidden layers and a Sigmoid activation at the output. The VAE is trained for 5 epochs using the Adam optimizer with a learning rate of 10^{-3}. After the VAE converges, we integrate and train three CouplingFlow layers as the NF module. Each flow permutes the

latent representation, splits it in half, and employs a small network with layers sized from half the dimension to 256, followed by another layer from 256 to 256, to produce scale and translation parameters for one half conditioned on the other. A learnable clamping mechanism ensures numerical stability in the scaling factors. Training the NF module for an additional 5 epochs with the Adam optimizer at a learning rate of 10^{-3} further refines the latent distribution. This two-step, disjoint training strategy ensures a more robust and expressive latent representation. To ensure a fair comparison, our model and SOTA models were assessed using the same number of samples, including ID, OOD, and on-manifold adversarial attacks (Section 5.4). This consistency was maintained either by implementing published codes, when available, or by reconstructing the code for each model based on the descriptions found in their respective papers. We use the normal network behavior from the KDD99 (Tavallaee et al., 2009) and X-IIoTID (Al-Hawawreh et al., 2022) datasets as ID samples, while treating the network attack behavior as OOD. Additionally, we employ an on-manifold attack approach that utilizes optimization and generative perturbation to create OOD samples with ID features.

5.1. Datasets

The X-IIoTID dataset (Al-Hawawreh et al., 2022) is a comprehensive collection for X-IIoTID security research, encompassing 820,834 samples across 68 features and covering a range of devices and traffic from edge to cloud. It includes detailed scenarios of both normal operations and various attacks such as probing, botnet activities, data exfiltration, tampering, crypto-ransomware, and Denial of Service (DoS). This dataset is crucial for analyzing and mitigating security vulnerabilities in X-IIoTID systems. The KDD99 dataset (Tavallaee et al., 2009) is a classic benchmark in the field of network security, used primarily for the evaluation of intrusion detection systems. It contains approximately 4.9 million network interactions, which are labeled as either normal (ID) or one of four main types of attacks (OOD) : DoS, probing, remote to local (infiltration attack), and user to root (privilege escalation attack) . This dataset helps researchers and security professionals in developing and testing the efficacy of their intrusion detection algorithms.

5.2. Preprocessing of ID and ODD samples

Before training any ML model, we perform extensive preprocessing on the network traffic data to ensure high-quality input for subsequent analysis. First, we used the extracted features from log and PCAP files in both datasets. Duplicate flows are then removed to eliminate redundancy, and missing values are imputed using the median to maintain robust statistical estimates. Next, the data is normalized to standardize the feature scales. Finally, the dataset is split into 80% for training and 20% for testing, with a strict isolation of testing samples to avoid data snooping issues.

5.3. Defense

In this section, we introduce a set of methods for comparison with our approach. These methods have been selected due to their SOTA performance in the cybersecurity domain or their role as baselines in adversarial robustness. Our evaluation incorporates both supervised and semi-supervised strategies and includes the following approaches: Deep Neural Networks (DNN) and Deep Reinforcement Learning (DRL) (Al-Hawawreh et al., 2021; Caminero et al., 2019); Adversarial Training, where we consider methods such as on-manifold AT (Xiao et al., 2025) and other off-manifold methods such as Misclassification Aware adveRsarial Training (MART) and TRadeoff-inspired Adversarial DEfense (TRADES) (Wang et al., 2020; Zhang et al., 2019); Input Preprocessing (IP), which includes input transformations using DAE and Contrastive Denoising Autoencoder(CDAE) during both the training and testing phases for DNN and DRL models (Sethi et al., 2020; Al-Hawawreh et al., 2021), and Feature Squeezing (FS) (Xu et al., 2018), proven effective against adversarial attacks; AEs in Anomaly Detection (MagNet and DRM) (Meng and Chen, 2017; Al-Fawa'reh et al., 2025); and the NF approach in Rudolph et al. (2020) as a density-based method. Recent defenses have also been considered, including RAIDS (Sarıkaya et al., 2023), a reconstruction-based defense; Apollon (Paya et al., 2024), which employs a model selection strategy; and IDS-AntA (Barik and Misra, 2024), which integrates feature selection and model selection using the Ant Colony Optimization algorithm and uses TTTS (Cohen et al., 2024) as a post-preprocessing defense.

5.4. Attacks

To effectively generate on-manifold attacks that reflect realistic attack scenarios, we integrate methods from off-manifold algorithms and generative perturbations, similar to those described in Xiao et al. (2025) and Al-Fawa'reh et al. (2025). These attacks are crafted under a black-box setting where the attacker does not have access to the target IDS, thereby mimicking real-world constraints. The focus is on leveraging attack transferability properties, as DNN models often share similarities in architecture, such as neuron structures and activation functions.

For enhanced simulation fidelity, we use the same substitute models as in Al-Fawa'reh et al. (2025). Unlike off-manifold adversarial attacks, which aim to deceive classifiers f_θ by optimizing

$$x_{\text{adv}} = \underset{x'}{\text{argmax}}\ \mathcal{L}\big(F(x'), y_{\text{true}}\big)$$

within an epsilon-bounded perturbation norm $\|x'-x\|_p \le \epsilon$, our adapted on-manifold approach adds constraints to ensure that perturbations not only stay within this bound but also adhere to the data manifold $\mathcal{M}$. This approach is formalized as

$$x_{\text{on-adv}} = \underset{x'}{\text{argmax}}\ \mathcal{L}\big(F(x'), y_{\text{true}}\big),$$

subject to $\|x' - x\|_p \le \epsilon$ and $x' \in \mathcal{M}$. By preserving the intrinsic characteristics and original labels of the data, this strategy makes it harder for defense models to detect these complex adversarial attacks, as typical data features remain intact. The constraints targeted the non-functional characteristics of the ODD samples, including Duration, total bytes, total packet, packet rate, and byte rate.

The adapted approach to generating adversarial attacks can be summarized as follows. First, we use existing algorithms like BIM, C&W, and DF in the high-dimensional feature space to create off-manifold adversarial attacks (Steps 1 & 3 in Fig. 4). Next, we utilize a Saliency Adversarial Autoencoder (SAAE) to convert these off-manifold samples into on-manifold ones that mimic normal network behavior (Steps 4–8 in Fig. 4). This phase employs a two-stage process. Initially, a standard autoencoder is trained on ID samples (Step 2 in Fig. 4). The off-manifold adversarial samples are then encoded into latent vectors and fused with a Gaussian distribution (Step 4 in Fig. 4), and a GAN-like structure refines these vectors by aligning them more closely with the latent distribution of ID traffic (Steps 7 & 8 in Fig. 4). Finally, the refined vectors are decoded back into the original high-dimensional feature space and then projected into that space (Step 9 in Fig. 4), producing on-manifold adversarial samples. As a result, the generated attacks not only retain the structural integrity of legitimate ID network traffic but also evade detection more effectively. Modified adversarial attacks (OBIM, ODF, OCWL_∞) constrained to the data manifold present unique cybersecurity challenges. These implications can be summarized as follows:

- **Stealth and Evasion:** By perturbing features within the natural traffic distribution (e.g., adjusting packet sizes or timing within valid protocol ranges), these attacks evade rule-based and statistical detectors. For example, adversarial DDoS traffic that mimics legitimate bursts can bypass threshold-based detection.

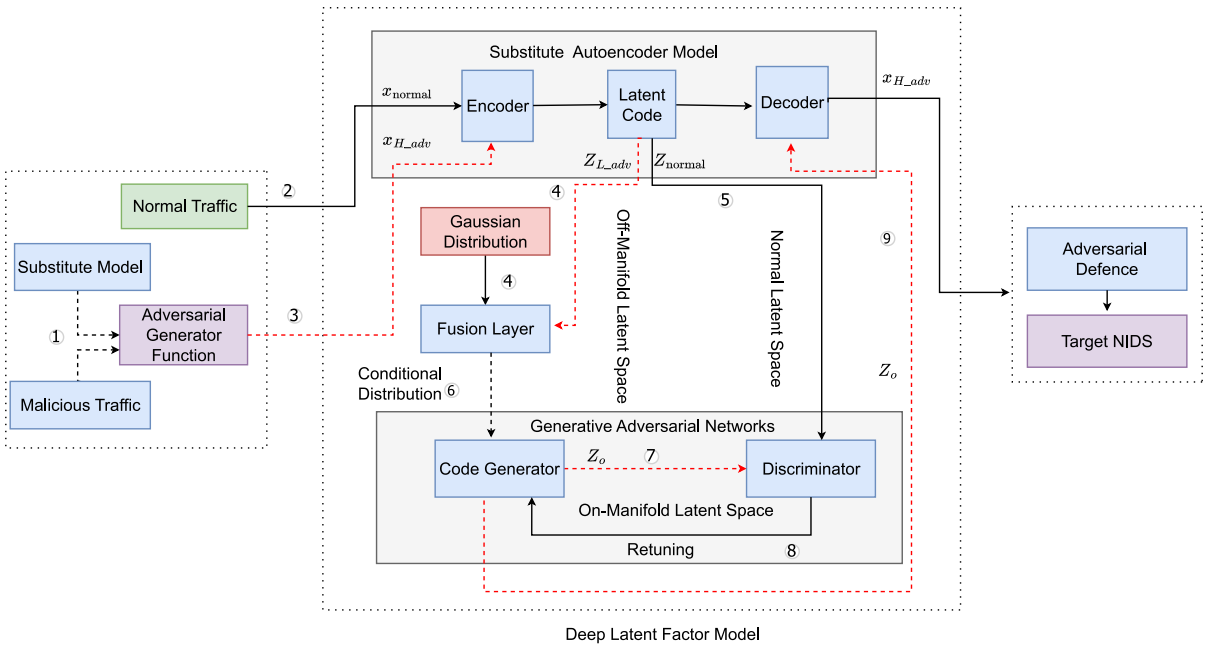

Fig. 4. On-manifold attacks generation process using SAAE Model (Al-Fawa'reh et al., 2025).

- **Defense Limitations:** SOTA defenses (as covered in Section 5.3), such as input preprocessing and off-manifold adversarial training, often fail because on-manifold attacks exploit subtle intrinsic feature correlations present in the training data that classifiers inadequately model.
- **Inherent Vulnerability:** Since these attacks conform to valid traffic distributions, even models trained on recent attack data remain susceptible.

Across the evaluated models based on the X-IIoTID network environment, the introduction of on-manifold adversarial attacks results in substantial degradation in detection performance. For example, the DNN model's detection rate drops by roughly 50% (from about 99.7% on clean data to approximately 50% under OBIM, ODF, and OCW attacks). Similarly, the DNN-CDAE model experiences a degradation ranging between 30% and 39%, and the RAIDS model shows a drop of nearly 48% under OBIM/ODF conditions. In stark contrast, the DNN-DAE model's performance collapses, with its detection rate falling from around 98% on clean data to merely about 1% under adversarial conditions—a near-complete loss of detection capability. While some DRL-based models show more moderate reductions (ranging from 3% to 20% depending on the attack type), these overall results clearly indicate that even SOTA defenses can be severely compromised by on-manifold adversarial perturbations.

6. Results and discussion

6.1. Results

This section evaluates our model's OOD generalization and robustness using the KDD99 (Tavallaee et al., 2009) and X-IIoTID (Al-Hawawreh et al., 2022) datasets. Our results, presented in Tables 2, 3, 4, and 5 (with the highest scores highlighted in **bold**), provide a comprehensive comparison of our model against various defense mechanisms under both no adversarial defense (No-Adv) and on-manifold attack scenarios.

Figs. 5 and 6 present the average scores for different defense methods on the KDD99 and X-IIoTID datasets, highlighting variations in performance under both clean and adversarial conditions. In the absence of adversarial attacks, the DNN model achieves the highest accuracy on the X-IIoTID dataset, whereas the EDLT model yields the best accuracy on the KDD99 dataset, as shown in Table 2. However, when adversarial attacks are introduced, the performance landscape shifts significantly. As observed in Table 5 based on the X-IIoTID dataset, our evaluation reveals that the average reduction in the F1 score across the evaluated models that employed supervised learning strategies is approximately 21.8%. Notably, the drop in F1 varied considerably among the models—while the ON_ADV model, which is trained on on-manifold samples, exhibited a minimal decrease of around 0.0156, the DNN-DAE model experienced a significant decline of 0.726. The EDLT model consistently outperforms other models, achieving the highest Accuracy and F1 scores on the KDD99 dataset, as shown in Tables 2 and 5. A similar trend is observed for the X-IIoTID dataset, where the EDLT model demonstrates superior performance in two attack cases (OBIM and ODF) and secures the second-best score in the OCW L_∞ attack, as detailed in Tables 2 and 5. As illustrated in Figs. 7 and 8, the EDLT model maintains a consistent behavior across clean and adversarial attack scenarios, with similar values in both conditions. This suggests that the model achieves a satisfactory level of robustness and generalizability. However, as illustrated in Table 3, the EDLT model demonstrates lower precision compared to some supervised methods such as RAIDS and IDS-anta, which suggests a higher false positive rate. It is important to recognize that the supervised methods are trained on both ID and OOD samples, whereas our approach is exclusively trained on IDS samples. This discrepancy highlights an opportunity for further improvement in our method.

6.2. Discussion

As detailed in Section 6.1, EDLT outperforms existing defenses by leveraging latent space transformations and quantifying DE changes to effectively detect subtle, on-manifold adversarial perturbations. In contrast to conventional models that rely on static feature representations or reconstruction errors, our approach first projects network traffic data into a robust, low-dimensional manifold using a VAE. Invertible transformations via NF then create a transformed space where even small adversarial perturbations yield measurable shifts in DE. Since DE quantifies the uncertainty and spread of the latent distribution, any perturbation, even if it lies on the data manifold, results in a notable entropy shift. This focus on DE change provides a critical advantage, as

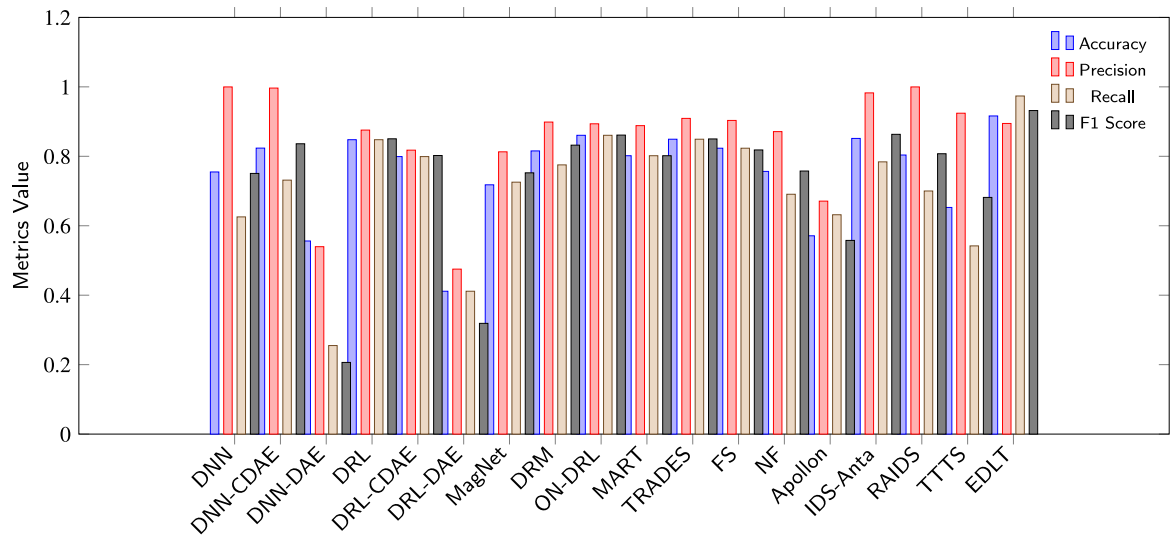

Fig. 5. Average Performance of SOTA Defenses vs. EDLT on the KDD99 dataset.

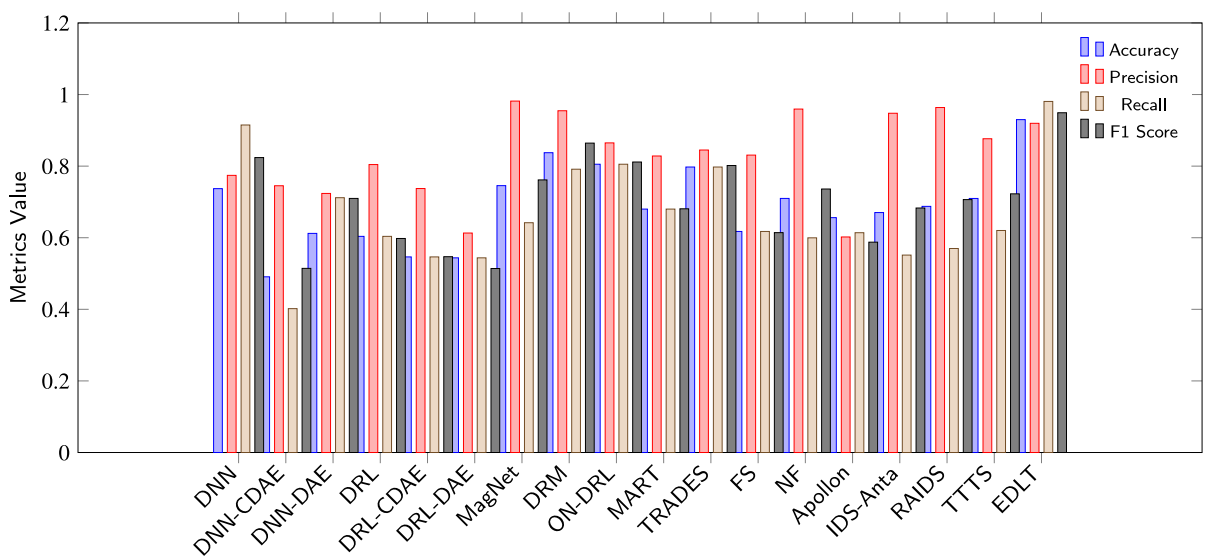

Fig. 6. Average performance of SOTA Defenses vs. EDLT on the X-IIoTI dataset.

Table 2
Accuracy scores on KDD99 and X-IIoTID datasets.

Model	Clean		OBIM		ODF		OCW L_∞	
	KDD99	X-IIoTID	KDD99	X-IIoTID	KDD99	X-IIoTID	KDD99	X-IIoTID
DNN (Al-Hawawreh et al., 2021)	0.7837	**0.9984**	0.7215	0.6716	0.7216	0.6739	0.7216	0.6755
DNN-CDAE (Al-Hawawreh et al., 2021)	0.6233	0.9855	0.4446	0.7959	0.4885	0.7345	0.4067	0.7786
DNN-DAE (Sethi et al., 2020)	0.6783	0.6830	0.5382	0.5138	0.7110	0.5138	0.5210	0.5138
DRL (Caminero et al., 2019)	0.7123	0.9224	0.4805	0.8477	0.4576	0.7328	0.7646	0.8882
DRL-CDAE (Al-Hawawreh et al., 2021)	0.7146	0.8672	0.4805	0.8175	0.4645	0.7213	0.5261	0.7909
DRL-DAE (Sethi et al., 2020)	0.4750	0.4955	0.4980	0.4955	0.7018	0.3273	0.5005	0.3273
MagNet (Meng and Chen, 2017)	0.7905	0.7942	0.7171	0.6728	0.7315	0.6411	0.7431	0.7637
DRM (Al-Fawa'reh et al., 2025)	0.7777	0.7774	0.8582	0.8499	0.8552	0.8078	0.8590	0.8270
ON-DRL (Xiao et al., 2025)	0.7684	0.8745	0.8081	0.9230	0.8062	0.7938	0.8387	0.8509
MART (Wang et al., 2020)	0.7633	0.9900	0.7233	0.6733	0.5933	0.7300	0.6400	0.8133
TRADES (Zhang et al., 2019)	0.7233	0.9900	0.8233	0.7233	0.8167	0.7567	0.8267	**0.9267**
FS (Xu et al., 2018)	0.6567	0.9600	0.6300	0.7967	0.5733	0.7300	0.6100	0.8067
NF (Rudolph et al., 2020)	0.7507	0.6957	0.6960	0.7911	0.6963	0.7520	0.6960	0.7872
Apollon (Paya et al., 2024)	0.7066	0.7889	0.6635	0.5116	0.6267	0.5070	0.6277	0.4768
IDS-Anta (Barik and Misra, 2024)	0.7690	0.9634	0.6551	0.8213	0.5829	0.8048	0.6740	0.8165
RAIDS (Sarıkaya et al., 2023)	0.7839	0.9992	0.5873	0.6870	0.5766	0.6789	0.8028	0.8496
TTTS (Cohen et al., 2024)	0.7369	0.8029	0.6013	0.6480	0.6401	0.6731	0.6401	0.7147
EDLT	**0.9037**	0.8761	**0.9388**	**0.9356**	**0.9385**	**0.9347**	**0.9387**	0.9185

Table 3
Precision scores on KDD99 and X-IIoTID datasets.

Model	Clean		OBIM		ODF		OCW L_∞	
	KDD99	X-IIoTID	KDD99	X-IIoTID	KDD99	X-IIoTID	KDD99	X-IIoTID
DNN (Al-Hawawreh et al., 2021)	0.9237	**0.9998**	0.7244	**0.9998**	0.7244	**0.9998**	0.7244	0.9998
DNN-CDAE (Al-Hawawreh et al., 2021)	0.7155	0.9957	0.7561	0.9969	0.7846	0.9965	0.7247	0.9968
DNN-DAE (Sethi et al., 2020)	0.6649	0.6079	0.7249	0.5170	0.7880	0.5170	0.7170	0.5170
DRL (Caminero et al., 2019)	0.8179	0.9225	0.7266	0.8706	0.8046	0.8113	0.8685	0.8977
DRL-CDAE (Al-Hawawreh et al., 2021)	0.7583	0.8754	0.7266	0.8255	0.7191	0.7633	0.7455	0.8065
DRL-DAE (Sethi et al., 2020)	0.6408	0.5025	0.5565	0.5025	0.6850	0.4477	0.5700	0.4477
MagNet (Meng and Chen, 2017)	**0.9717**	0.7698	**0.9837**	0.8169	**0.9854**	0.8120	**0.9858**	0.8528
DRM (Al-Fawa'reh et al., 2025)	0.9198	0.8311	0.9664	0.9239	0.9661	0.9183	0.9663	0.9208
ON-DRL (Xiao et al., 2025)	0.8268	0.8873	0.8735	0.9279	0.8730	0.8661	0.8864	0.8933
MART (Wang et al., 2020)	0.8233	0.9900	0.8467	0.8300	0.8167	0.8467	0.8267	0.8867
TRADES (Zhang et al., 2019)	0.7767	0.9900	0.8700	0.8467	0.8633	0.8567	0.8700	0.9433
FS (Xu et al., 2018)	0.8067	0.9633	0.8433	0.8933	0.8333	0.8667	0.8400	0.8900
NF (Rudolph et al., 2020)	0.9370	0.8825	0.9668	0.8709	0.9668	0.8607	0.9668	0.8701
Apollon (Paya et al., 2024)	0.7307	0.8118	0.6272	0.6383	0.5079	0.6304	0.5425	0.6031
IDS-Anta (Barik and Misra, 2024)	0.9347	0.9737	0.9578	0.9864	0.9433	0.9848	0.9549	0.9855
RAIDS (Sarıkaya et al., 2023)	0.9517	**0.9998**	0.9644	**0.9998**	0.9604	**0.9998**	0.9783	**0.9999**
TTTS (Cohen et al., 2024)	0.8929	0.8369	0.9309	0.8815	0.9376	0.8887	0.9350	0.8989
EDLT	0.8752	0.8120	0.9345	0.9227	0.9343	0.9227	0.9345	0.9209

Table 4
Recall scores on KDD99 and X-IIoTID datasets.

Model	Clean		OBIM		ODF		OCW L_∞	
	KDD99	X-IIoTID	KDD99	X-IIoTID	KDD99	X-IIoTID	KDD99	X-IIoTID
DNN (Al-Hawawreh et al., 2021)	0.6759	**0.9970**	**0.9945**	0.4985	**0.9945**	0.5020	**0.9945**	0.5044
DNN-CDAE (Al-Hawawreh et al., 2021)	0.5614	0.9745	0.3460	0.6904	0.4065	0.5966	0.2938	0.6639
DNN-DAE (Sethi et al., 2020)	0.8766	0.9817	0.5859	0.0128	0.8232	0.0128	0.5613	0.0128
DRL (Caminero et al., 2019)	0.7123	0.9224	0.4805	0.8477	0.4576	0.7328	0.7646	0.8882
DRL-CDAE (Al-Hawawreh et al., 2021)	0.7146	0.8672	0.4805	0.8175	0.4645	0.7213	0.5261	0.7909
DRL-DAE (Sethi et al., 2020)	0.4750	0.4955	0.4980	0.4955	0.7018	0.3273	0.5005	0.3273
MagNet (Meng and Chen, 2017)	0.6519	0.8734	0.6200	0.6483	0.6399	0.5929	0.6559	0.7872
DRM (Al-Fawa'reh et al., 2025)	0.6677	0.6810	0.8336	0.8398	0.8295	0.7754	0.8348	0.8050
ON-DRL (Xiao et al., 2025)	0.7684	0.8745	0.8081	0.9230	0.8062	0.7938	0.8387	0.8509
MART (Wang et al., 2020)	0.7633	0.9900	0.7233	0.6733	0.5933	0.7300	0.6400	0.8133
TRADES (Zhang et al., 2019)	0.7233	0.9900	0.8233	0.7233	0.8167	0.7567	0.8267	0.9267
FS (Xu et al., 2018)	0.6567	0.9600	0.6300	0.7967	0.5733	0.7300	0.6100	0.8067
NF (Rudolph et al., 2020)	0.5995	0.4286	0.5996	0.7999	0.5999	0.7402	0.5996	0.7939
Apollon (Paya et al., 2024)	0.7204	0.7853	0.6411	0.5913	0.5305	0.5868	0.5641	0.5633
IDS-Anta (Barik and Misra, 2024)	0.6388	0.9499	0.5469	0.7396	0.4474	0.7144	0.5728	0.7322
RAIDS (Sarıkaya et al., 2023)	0.6537	0.9985	0.4478	0.5220	0.4331	0.5096	0.7449	0.7703
TTTS (Cohen et al., 2024)	0.6113	0.7389	0.4865	0.5342	0.5399	0.5724	0.5305	0.6354
EDLT	0.9690	0.9701	0.9846	**0.9841**	0.9845	**0.9826**	0.9845	**0.9579**

Table 5
F1 scores on KDD99 and X-IIoTID datasets.

Model	Clean		OBIM		ODF		OCW L_∞	
	KDD99	X-IIoTID	KDD99	X-IIoTID	KDD99	X-IIoTID	KDD99	X-IIoTID
DNN (Al-Hawawreh et al., 2021)	0.7806	0.9984	0.8382	0.6653	0.8383	0.6684	0.8383	0.6705
DNN-CDAE (Al-Hawawreh et al., 2021)	0.6292	0.9850	0.4748	0.8158	0.5356	0.7464	0.4181	0.7970
DNN-DAE (Sethi et al., 2020)	0.7562	0.7509	0.6480	0.0249	0.8052	0.0249	0.6296	0.0249
DRL (Caminero et al., 2019)	0.7010	0.9224	0.4803	0.8510	0.4329	0.7383	0.7772	0.8899
DRL-CDAE (Al-Hawawreh et al., 2021)	0.7125	0.8667	0.4803	0.8198	0.4603	0.7281	0.5347	0.7946
DRL-DAE (Sethi et al., 2020)	0.3183	0.4748	0.5208	0.4748	0.6916	0.1629	0.5253	0.1629
MagNet (Meng and Chen, 2017)	0.7789	0.8077	0.7595	0.7076	0.7493	0.6799	0.7583	0.8144
DRM (Al-Fawa'reh et al., 2025)	0.7737	0.7486	0.8951	0.8799	0.8926	0.8409	0.8958	0.8590
ON-DRL (Xiao et al., 2025)	0.7654	0.8727	0.8181	0.9236	0.8162	0.7962	0.8464	0.8516
MART (Wang et al., 2020)	0.7567	0.9900	0.7367	0.6700	0.5900	0.7300	0.6400	0.8167
TRADES (Zhang et al., 2019)	0.7233	0.9900	0.8300	0.7233	0.8233	0.7567	0.8300	0.9300
FS (Xu et al., 2018)	0.6333	0.9600	0.6367	0.7867	0.5700	0.7200	0.6167	0.8067
NF (Rudolph et al., 2020)	0.7299	0.5713	0.7381	0.8337	0.7384	0.7953	0.7381	0.8301
Apollon (Paya et al., 2024)	0.7038	0.7837	0.6202	0.4965	0.5048	0.4924	0.5218	0.4585
IDS-Anta (Barik and Misra, 2024)	0.7588	0.9616	0.6866	0.8363	0.5899	0.8214	0.6967	0.8337
RAIDS (Sarıkaya et al., 2023)	0.7750	**0.9991**	0.6114	0.6859	0.5939	0.6751	0.8456	0.8696
TTTS (Cohen et al., 2024)	0.7257	0.7849	0.6389	0.6652	0.6850	0.6963	0.6766	0.7441
EDLT	**0.9197**	0.8840	**0.9589**	**0.9524**	**0.9587**	**0.9517**	**0.9588**	**0.9390**

it enables our model to flag even the most subtle intrusions. Our experimental evaluation on the KDD99 and X-IIoTID cybersecurity datasets demonstrates that under adversarial conditions, EDLT attains recall values of approximately 97%, outperforming competing defenses by 15 to 30%. Notably, the recall performance on both clean and adversarial samples underscores the inherent robustness and generalizability of our approach. On clean data, the model already exhibits high recall, ensuring that ID network traffic is correctly recognized; when adversarial

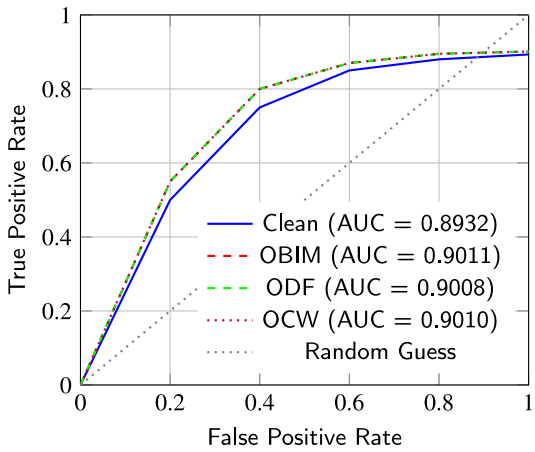

Fig. 7. AUC-ROC curves for KDD99 dataset.

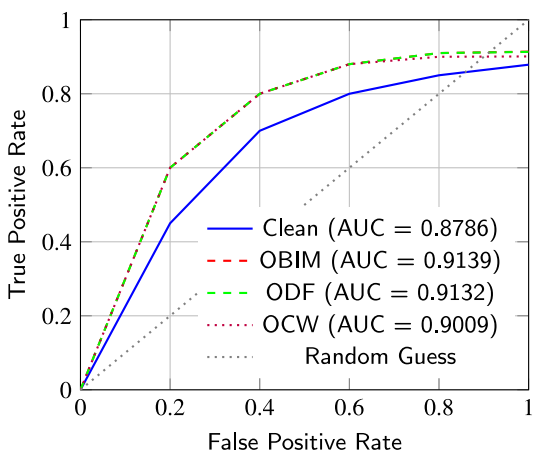

Fig. 8. AUC-ROC curves for X-IIoTID dataset.

perturbations are introduced, the consistently high recall confirms that the intrinsic structure of ID traffic is preserved even under attack. This stability is crucial in cybersecurity, where missing a single intrusion (false negative alert) can have severe consequences. A key advantage of EDLT is that it relies solely on ID samples rather than requiring both ID and OOD samples. This design choice is particularly beneficial in domains such as IDS, where obtaining OOD samples in real-world settings can be challenging. Moreover, since ID data is more readily available, updating the model becomes much simpler. Our evaluation further reinforces this benefit: while alternative methods that depend exclusively on ID samples have proven less reliable under on-manifold attacks, EDLT maintains robust performance even when confronted with such adversarial conditions. Another advantage of EDLT is its use of secret knowledge, applying unique permutations before every transformation phase. Because these transformations remain confidential, they create a fundamental barrier to attackers. To succeed, adversaries must approximate a transformed manifold, a challenge that forces them to introduce perturbations likely to trigger detectable DE shifts. In practical terms, this secret knowledge not only enhances the system's ability to detect anomalies but also significantly raises the barrier against sophisticated intrusions, ensuring that even subtle, on-manifold adversarial tactics are reliably identified. In a broader sense, the EDLT approach has strong potential for securing critical infrastructure. For example, in an industrial control system at a smart manufacturing facility, sensors and actuators typically generate consistent ID data (such as temperature, pressure, or vibration readings). An attacker could introduce subtle, on-manifold perturbations—small alterations that keep the data statistically plausible yet trigger dangerous operational decisions. Using EDLT, security analysts can first learn a safe profile of normal ICS traffic based on ID samples. Incoming data is then assessed against this learned profile, flagging subtle deviations that might bypass conventional IDS. By catching stealthy attacks early, EDLT helps prevent major disruptions and proves well-suited for real-time deployment in critical industrial operations.

7. Computational efficiency of adversarial defense models

Fig. 9 compares the training and testing times of various adversarial defense models on a system equipped with an Intel® Core i7-6500U CPU @ 2.50 GHz–2.60 GHz, 16.0 GB of memory, and a 64-bit operating system. While EDLT does not boast the fastest training and testing times, it exhibits moderate yet highly effective computational efficiency in detection tasks. Notably, each preprocessed sample requires approximately 0.0057 s (5.7 ms) for training and only about 0.000063 s (0.063 ms) for testing, underscoring its suitability for real-time applications. When benchmarked against existing network security solutions like next-generation firewalls and IDS systems, EDLT shows considerable potential for online deployment. This capability suggests that our model can be seamlessly integrated into live security infrastructures to deliver prompt adversarial detection in dynamic network environments. Future work may explore techniques such as model distillation, where a pre-trained teacher model guides the training of a more efficient student model, and the offline precomputation of intensive operations to further reduce runtime overhead.

8. Limitations

While our study provides valuable insights into latent space transformations and their relationship to DE under adversarial settings, several limitations must be noted. First, the assumption that key-driven transformations and internal states remain concealed — thereby providing a strategic advantage to the defender — may be problematic if the model design is made fully open to align with the principles of open ML. In such scenarios, the robustness of the proposed transformations might need to be re-evaluated. Second, our model exhibited a lower precision rate compared to the other defenses (DNN, IDS-Anta, RAIDS, and TTTS), as shown in 3, leading to higher false positives—a potentially undesirable outcome if insufficient experts are available to verify these alerts in real-world deployments. Finally, it is important to carefully consider the number of transformations selected, as a larger number may increase the training time and computational burden. However, in this study, our model requires only three transformation flows, ensuring a lightweight training process while maintaining efficiency and robustness.

9. Future directions

Future research should address the limitations outlined above to further enhance the robustness and applicability of our approach. One promising direction is to develop mechanisms that preserve security while supporting transparency in open ML systems, thereby mitigating issues arising from the need to keep key-driven transformations and internal states concealed. To address the precision issue, we will explore more extensive hyperparameter tuning and consider incorporating a manifold-aware loss function. Furthermore, future work will focus on integrating defenses against advanced adversarial techniques to ensure comprehensive protection of the model.

10. Ablation study

- In this study, we investigated the impact of the clamping mechanism on the dynamic transformations employed by our model.

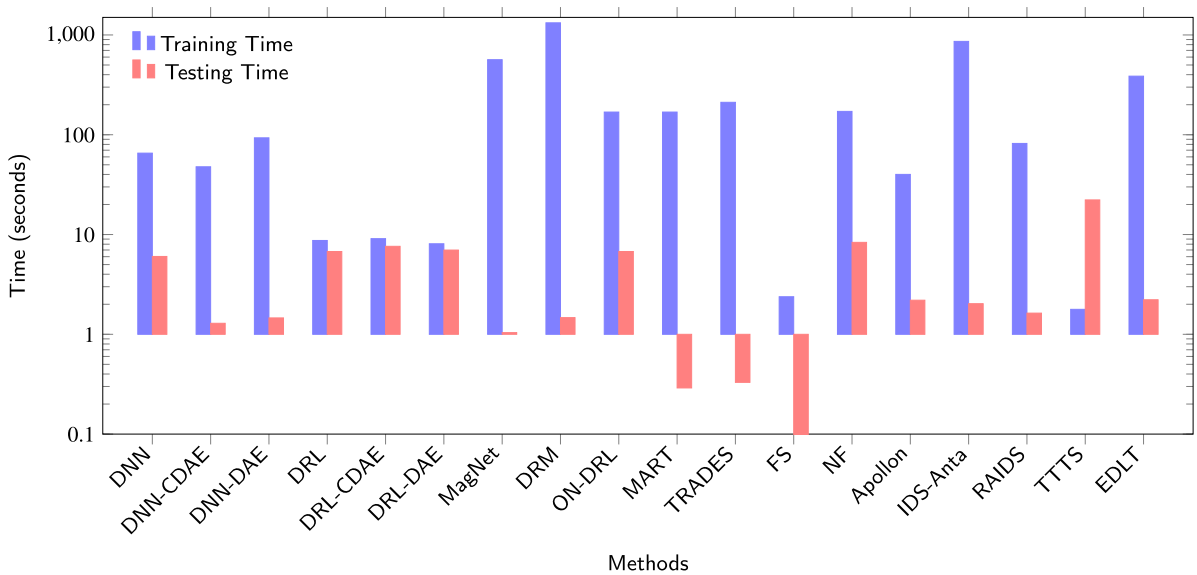

Fig. 9. Training and testing times for various adversarial defense methods (displayed on a logarithmic scale).

Table 6
Clamping vs. Non-Clamping evaluation.

Method	Clean	OBIM	ODF	OCW L_∞
KDD dataset				
Clamp	**0.8932**	**0.901**	**0.901**	**0.901**
No clamp	0.500	0.500	0.500	0.500
X-IIoTID dataset				
Clamp	**0.8786**	**0.914**	**0.913**	**0.909**
No clamp	0.875	0.891	0.891	0.888

Our analysis revealed that the scaling operations within the affine coupling layers can produce extremely large or small values, which in turn may lead to exploding or vanishing gradients and destabilize the training process. For example, our ablation study demonstrated that for the KDD99 dataset, employing clamping maintained stable AUC-ROC values — approximately 0.893 for clean inputs and 0.901 for adversarial inputs — whereas the absence of clamping resulted in a uniform drop to 0.5 (see Table 6). Although the performance difference on the X-IIoTID dataset was more marginal, clamping still yielded consistently superior results, reinforcing its importance in managing numerical instability. By allowing the clamping limits to be adjusted based on the data distribution during training, we can mitigate extreme value propagation.

- To further assess our model's generalization capabilities and to demonstrate its resilience against data snooping, we conducted an ablation study using a variant of the Jacobian-based Saliency Map Attack (JSMA) adapted to remain on the data manifold (referred to as OJSMA) (Al-Fawa'reh et al., 2025). Notably, these attack samples were unseen during training. By comparing the model's performance on clean inputs versus OJSMA-perturbed samples for both the X-IIoTID and KDD99 datasets, we emulate a scenario in which attackers subtly adjust non-functional features (e.g., inter-packet timing, header attributes) while adhering to legitimate network protocols.

  As shown in Table 7, the model maintains high AUC-ROC scores even when faced with on-manifold JSMA modifications, indicating robust generalization to stealthy threats. Notably, the scores for OJSMA are slightly higher than those for clean data, which may stem from the latent-space transformations capturing subtle artifacts of the adversarial manipulation. From a data snooping standpoint, these results confirm that our strict train/test split strategy (Section 3.3) prevents inadvertently tuning the model to specific adversarial patterns. Notably, our training approach exclusively utilizes ID samples, which inherently reduces the risk of data snooping. This methodology also offers a practical advantage for future scalability: it is relatively straightforward and cost-effective to label ID network traffic, whereas obtaining labeled OOD or attack samples is both expensive and unpredictable—typically only feasible if a company experiences an actual attack. Consequently, the model remains effective in identifying previously unseen, on-manifold perturbations, and the differential entropy shifts in the transformed latent space reliably detect OOD and adversarial samples without leaking future attack information into the training process (see Table 8).

Table 7
AUC-ROC Comparison of clean vs. OJSMA Attack.

	Clean	OJSMA
KDD99	0.893	0.9010
X-IIoTID	0.879	0.9038

11. Conclusion

This paper addresses the critical challenges of OOD generalization and robustness in the cybersecurity domain. The work demonstrates the limitations of existing detection methods, which fail to achieve satisfactory levels of generalizability and robustness, particularly against on-manifold adversarial attacks. To overcome these challenges, we propose EDLT, a new method that relies on changes in the DE of the latent space of ID samples during multiple transformations as a baseline for identifying OOD samples. Experimental results validate the effectiveness of this approach across diverse environments and various on-manifold adversarial attack scenarios. These findings highlight the robustness and generalizability of the EDLT model, showcasing its potential applicability in a wide range of security-sensitive settings.

Table 8
Table of abbreviations.

Abbreviation	Definition
AE	Autoencoder
Apollon	OOD detection via dynamic classifier selection (using Multi-Armed Bandits)
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
BIM	Basic Iterative Method (adversarial attack)
CDAE	Contrastive Denoising Autoencoder
CW L_p	Carlini & Wagner Attack (with L_p norm)
DAE	Denoising Autoencoder
DE	Differential Entropy
DF	DeepFool (adversarial attack)
DNN	Deep Neural Network
DNN-CDAE	DNN with Contrastive Denoising Autoencoder
DNN-DAE	DNN with Denoising Autoencoder
DRL	Deep Reinforcement Learning
DRL-CDAE	DRL with Contrastive Denoising Autoencoder
DRL-DAE	DRL with Denoising Autoencoder
DRM	Data Refinement Module
DS	Deviation Score
EDLT	Entropy-Driven Latent Transformation
FS	Feature Squeezing
FN	False Negative
FP	False Positive
GAN	Generative Adversarial Network
ID	In-Distribution (normal data, legitimate traffic)
IDS-Anta	Intrusion Detection System using Ant Colony Optimization
JSMA	Jacobian-based Saliency Map Attack
KDD99	KDD 1999 Dataset
ML	Machine Learning
MAB	Multi-Armed Bandits
MART	Misclassification Aware adversarial Training
NF	Normalizing Flow
NIDS	Network Intrusion Detection System
No-Adv	No Adversarial Defense
OBIM	On-manifold BIM
ODF	On-manifold DeepFool
OCW L_∞	On-manifold Carlini & Wagner Attack with L_∞ norm
OOD	Out-of-Distribution
OJSMA	On-manifold Jacobian-based Saliency Map Attack
PGD	Projected Gradient Descent
RAIDS	Robust Adversarial Intrusion Detection System
RE	Reconstruction Error
SOTA	State-Of-The-Art
TRADES	TRadeoff-inspired Adversarial DEfense
TTTS	Tree Test Time Simulation
VAE	Variational Autoencoder
X-IIoTID	Extended Industrial Internet of Things Intrusion Dataset

CRediT authorship contribution statement

Mohmmad Al-Fawa'reh: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jumana Abu-khalaf:** Writing – review & editing, Supervision, Resources, Investigation. **Naeem Janjua:** Writing – review & editing, Supervision. **Patryk Szewczyk:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

Adeke, J.M., Liu, G., Amoah, L., Nwali, O.J., 2025. Investigating the impact of feature selection on adversarial transferability in intrusion detection system. Comput. Secur. 151, 104327. http://dx.doi.org/10.1016/j.cose.2025.104327, URL https://www.sciencedirect.com/science/article/pii/S0167404825000161.

Aiken, J., Scott-Hayward, S., 2019. Investigating adversarial attacks against network intrusion detection systems in SDNs. In: 2019 IEEE Conference on Network Function Virtualization and Software Defined Networks (NFV-SDN). pp. 1–7. http://dx.doi.org/10.1109/NFV-SDN47374.2019.9040101.

Al-Fawa'reh, M., Abu-khalaf, J., Janjua, N., Szewczyk, P., 2025. On and off the manifold: Generation and detection of adversarial attacks in IIoT networks. J. Netw. Comput. Appl. 235, 104102. http://dx.doi.org/10.1016/j.jnca.2024.104102, URL https://www.sciencedirect.com/science/article/pii/S1084804524002790.

Al-Hawawreh, M., Sitnikova, E., Aboutorab, N., 2021. Asynchronous peer-to-peer federated capability-based targeted ransomware detection model for industrial IoT. IEEE Access 9, 148738–148755. http://dx.doi.org/10.1109/ACCESS.2021.3124634.

Al-Hawawreh, M., Sitnikova, E., Aboutorab, N., 2022. X-IIoTID: A connectivity-agnostic and device-agnostic intrusion data set for industrial internet of things. IEEE Internet Things J. 9 (5), 3962–3977. http://dx.doi.org/10.1109/JIOT.2021.3102056.

Amirullah, A., Aulia, I., Arisandy, D., 2020. Implementing cosine similarity algorithm to increase the flexibility of hematology text report generation. In: 2020 International Conference on Data Science, Artificial Intelligence, and Business Analytics. DATABIA, pp. 76–82. http://dx.doi.org/10.1109/DATABIA50434.2020.9190549.

AprilPyone, M., Kiya, H., 2020. An extension of encryption-inspired adversarial defense with secret keys against adversarial examples. In: 2020 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC). pp. 1369–1374.

Barik, K., Misra, S., 2024. IDS-Anta: An open-source code with a defense mechanism to detect adversarial attacks for intrusion detection system. Softw. Impacts 21, 100664. http://dx.doi.org/10.1016/j.simpa.2024.100664, URL https://www.sciencedirect.com/science/article/pii/S2665963824000526.

Bogun, A.A., Kostadinov, D., Borth, D., 2021. Saliency diversified deep ensemble for robustness to adversaries. ArXiv abs/2112.03615 URL https://api.semanticscholar.org/CorpusID:244920714.

Caminero, G., Lopez-Martin, M., Carro, B., 2019. Adversarial environment reinforcement learning algorithm for intrusion detection. Comput. Netw. 159, 96–109. http://dx.doi.org/10.1016/j.comnet.2019.05.013, URL https://www.sciencedirect.com/science/article/pii/S1389128618311216.

Carlini, N., Wagner, D.A., 2016. Towards evaluating the robustness of neural networks. In: 2017 IEEE Symposium on Security and Privacy (SP). pp. 39–57, URL https://api.semanticscholar.org/CorpusID:2893830.

Cho, M., Kim, T., Kim, W., Cho, S., Lee, S., 2020. Unsupervised video anomaly detection via normalizing flows with implicit latent features. Pattern Recognit. 129, 108703, URL https://api.semanticscholar.org/CorpusID:248117132.

Cohen, S., Arbili, O., Mirsky, Y., Rokach, L., 2024. TTTS: tree test time simulation for enhancing decision tree robustness against adversarial examples. In: Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence and Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence and Fourteenth Symposium on Educational Advances in Artificial Intelligence. In: AAAI'24/IAAI'24/EAAI'24, AAAI Press, http://dx.doi.org/10.1609/aaai.v38i19.30090.

Cook, E.D., Lavoie, M.-A., Waslander, S.L., 2024. Feature density estimation for out-of-distribution detection via normalizing flows. ArXiv abs/2402.06537 URL https://api.semanticscholar.org/CorpusID:267617280.

Corsini, A., Yang, S.J., 2023. Are existing out-of-distribution techniques suitable for network intrusion detection? In: 2023 IEEE Conference on Communications and Network Security (CNS). pp. 1–9, URL https://api.semanticscholar.org/CorpusID:261244119.

Dinh, L., Sohl-Dickstein, J.N., Bengio, S., 2016. Density estimation using real NVP. ArXiv abs/1605.08803 URL https://api.semanticscholar.org/CorpusID:8768364.

Doersch, C., 2016. Tutorial on variational autoencoders. ArXiv abs/1606.05908 URL https://api.semanticscholar.org/CorpusID:10510670.

Friedl, S., Nagel, M., Orson, P., Powell, M., 2019. A survey of the foundations of four-manifold theory in the topological category. ArXiv: Geom. Topol. URL https://api.semanticscholar.org/CorpusID:204734371.

Goodfellow, I.J., Shlens, J., Szegedy, C., 2014. Explaining and harnessing adversarial examples. CoRR abs/1412.6572 URL https://api.semanticscholar.org/CorpusID:6706414.

Goodge, A., Hooi, B., Ng, S.K., Ng, W.S., 2020. Robustness of autoencoders for anomaly detection under adversarial impact. In: Bessiere, C. (Ed.), Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, IJCAI-20. International Joint Conferences on Artificial Intelligence Organization, pp. 1244–1250. http://dx.doi.org/10.24963/ijcai.2020/173, URL https://doi.org/10.24963/ijcai.2020/173, Main track.

Gudovskiy, D., Ishizaka, S., Kozuka, K., 2022. CFLOW-AD: Real-Time unsupervised anomaly detection with localization via conditional normalizing flows. In: 2022 IEEE/CVF Winter Conference on Applications of Computer Vision. WACV, pp. 1819–1828. http://dx.doi.org/10.1109/WACV51458.2022.00188.

Hendrycks, D., Gimpel, K., 2016. A baseline for detecting misclassified and out-of-distribution examples in neural networks. ArXiv abs/1610.02136 URL https://api.semanticscholar.org/CorpusID:13046179.

Kingma, D.P., Dhariwal, P., 2018. Glow: Generative flow with invertible 1x1 convolutions. ArXiv abs/1807.03039 URL https://api.semanticscholar.org/CorpusID:49657329.

Kingma, D.P., Salimans, T., Jozefowicz, R., Chen, X., Sutskever, I., Welling, M., 2016. Improved variational inference with inverse autoregressive flow. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 4743–4751.

Kirichenko, P., Izmailov, P., Wilson, A.G., 2020. Why normalizing flows fail to detect out-of-distribution data. ArXiv abs/2006.08545 URL https://api.semanticscholar.org/CorpusID:219687356.

Kobyzev, I., Prince, S., Brubaker, M.A., 2019. Normalizing flows: Introduction and ideas. ArXiv abs/1908.09257 URL https://api.semanticscholar.org/CorpusID:201668500.

Kurakin, A., Goodfellow, I., Bengio, S., 2017. Adversarial examples in the physical world. arXiv:1607.02533 URL https://arxiv.org/abs/1607.02533.

Lee, K., Lee, K., Lee, H., Shin, J., 2018. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. ArXiv abs/1807.03888 URL https://api.semanticscholar.org/CorpusID:49667948.

Madry, A., Makelov, A., Schmidt, L., Tsipras, D., Vladu, A., 2017. Towards deep learning models resistant to adversarial attacks. ArXiv abs/1706.06083 URL https://api.semanticscholar.org/CorpusID:3488815.

Meng, D., Chen, H., 2017. MagNet: A two-pronged defense against adversarial examples. In: Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security. CCS '17, Association for Computing Machinery, New York, NY, USA, pp. 135–147. http://dx.doi.org/10.1145/3133956.3134057.

Moosavi-Dezfooli, S.-M., Fawzi, A., Frossard, P., 2015. DeepFool: A simple and accurate method to fool deep neural networks. In: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 2574–2582, URL https://api.semanticscholar.org/CorpusID:12387176.

Paya, A., Arroni, S., García-Díaz, V., Gómez, A., 2024. Apollon: A robust defense system against adversarial machine learning attacks in intrusion detection systems. Comput. Secur. 136, 103546. http://dx.doi.org/10.1016/j.cose.2023.103546, URL https://www.sciencedirect.com/science/article/pii/S016740482300456X.

Rezende, D.J., Mohamed, S., 2015. Variational inference with normalizing flows. In: Proceedings of the 32nd International Conference on Machine Learning. ICML, pp. 1530–1538.

Rudolph, M., Wandt, B., Rosenhahn, B., 2020. Same same but DifferNet: Semi-supervised defect detection with normalizing flows. In: 2021 IEEE Winter Conference on Applications of Computer Vision (WACV). pp. 1906–1915, URL https://api.semanticscholar.org/CorpusID:221370646.

Rudolph, M., Wehrbein, T., Rosenhahn, B., Wandt, B., 2021. Fully convolutional cross-scale-flows for image-based defect detection. In: 2022 IEEE/ CVF Winter Conference on Applications of Computer Vision (WACV). pp. 1829–1838, URL https://api.semanticscholar.org/CorpusID:238408276.

Samangouei, P., Kabkab, M., Chellappa, R., 2018. Defense-GAN: Protecting classifiers against adversarial attacks using generative models. ArXiv abs/1805.06605 URL https://api.semanticscholar.org/CorpusID:3458858.

Sarıkaya, A., Kılıç, B.G., Demirci, M., 2023. RAIDS: Robust autoencoder-based intrusion detection system model against adversarial attacks. Comput. Secur. 135, 103483. http://dx.doi.org/10.1016/j.cose.2023.103483, URL https://www.sciencedirect.com/science/article/pii/S0167404823003930.

Sethi, K., Rupesh, E.S., Kumar, R., Bera, P., Yatam, V.M., 2019. A context-aware robust intrusion detection system: a reinforcement learning-based approach. Int. J. Inf. Secur. 19, 657–678.

Sethi, K., Sai Rupesh, E., Kumar, R., Bera, P., Venu Madhav, Y., 2020. A Context-Aware Robust intrusion detection system: A reinforcement learning-based approach. Int. J. Inf. Secur. 19 (6), 657–678. http://dx.doi.org/10.1007/s10207-019-00482-7.

Shah, K., Chen, S., Klivans, A.R., 2023. Learning mixtures of Gaussians using the DDPM objective. ArXiv abs/2307.01178 URL https://api.semanticscholar.org/CorpusID:259316385.

Shannon, C.E., 1948. A mathematical theory of communication. Bell Syst. Tech. J. 27 (3), 379–423, 623–656. http://dx.doi.org/10.1002/j.1538-7305.1948.tb01338.x.

Stutz, D., Hein, M., Schiele, B., 2018. Disentangling adversarial robustness and generalization. In: 2019 IEEE/ CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 6969–6980, URL https://api.semanticscholar.org/CorpusID:54439061.

Sun, Y., Ming, Y., Zhu, X., Li, Y., 2022. Out-of-distribution detection with deep nearest neighbors. In: International Conference on Machine Learning. URL https://api.semanticscholar.org/CorpusID:248157551.

Taran, O., Rezaeifar, S., Voloshynovskiy, S., 2018. Bridging machine learning and cryptography in defence against adversarial attacks. arXiv:1809.01715 URL https://arxiv.org/abs/1809.01715.

Tarchoun, B., Khalifa, A.B., Mahjoub, M.A., Abu-Ghazaleh, N.B., Alouani, I., 2023. Jedi: Entropy-based localization and removal of adversarial patches. In: 2023 IEEE/ CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 4087–4095, URL https://api.semanticscholar.org/CorpusID:258236472.

Tavallaee, M., Bagheri, E., Lu, W., Ghorbani, A.A., 2009. A detailed analysis of the KDD CUP 99 data set. In: Proceedings of the Second IEEE Symposium on Computational Intelligence for Security and Defence Applications 2009. Ottawa, pp. 1–6. http://dx.doi.org/10.1109/CISDA.2009.5356528.

Wang, Y., Zou, D., Yi, J., Bailey, J., Ma, X., Gu, Q., 2020. Improving adversarial robustness requires revisiting misclassified examples. In: 8th International Conference on Learning Representations, ICLR 2020, Addis Ababa, Ethiopia, April 26-30, 2020. OpenReview.net, URL https://openreview.net/forum?id=rklOg6EFwS.

Xiao, J., Yang, L., Fan, Y., Wang, J., Luo, Z.-Q., 2025. Understanding adversarial robustness against on-manifold adversarial examples. Pattern Recognit. 159, 111071. http://dx.doi.org/10.1016/j.patcog.2024.111071, URL https://www.sciencedirect.com/science/article/pii/S0031320324008227.

Xu, W., Evans, D., Qi, Y., 2018. Feature squeezing: Detecting adversarial examples in deep neural networks. In: Proceedings 2018 Network and Distributed System Security Symposium. In: NDSS 2018, Internet Society, http://dx.doi.org/10.14722/ndss.2018.23198.

Yang, J., Zhou, K., Li, Y., Liu, Z., 2021. Generalized out-of-distribution detection: A survey. ArXiv abs/2110.11334 URL https://api.semanticscholar.org/CorpusID:239049401.

Zhang, C., Luo, X., Han, P., 2022. On-manifold adversarial attack based on latent space substitute model. Comput. Secur. 120, 102770. http://dx.doi.org/10.1016/j.cose.2022.102770, URL https://www.sciencedirect.com/science/article/pii/S0167404822001651.

Zhang, H., Yu, Y., Jiao, J., Xing, E.P., Ghaoui, L.E., Jordan, M.I., 2019. Theoretically principled trade-off between robustness and accuracy. abs/1901.08573 URL https://arxiv.org/abs/1901.08573.

Zhang, M., Zhang, A., McDonagh, S., 2021. On the Out-of-distribution generalization of probabilistic image modelling. In: Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., Vaughan, J.W. (Eds.), Advances in Neural Information Processing Systems. Vol. 34, Curran Associates, Inc., pp. 3811–3823.

Zou, X., Liu, W., 2023. On the adversarial robustness of out-of-distribution generalization models. In: Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., Levine, S. (Eds.), Advances in Neural Information Processing Systems. Vol. 36, Curran Associates, Inc., pp. 68908–68938, URL https://proceedings.neurips.cc/paper_files/paper/2023/file/d9888cc7baa04c2e44e8115588133515-Paper-Conference.pdf.

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